

MSc Data Science Project

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Department of Physics, Astronomy and Mathematics

Data Science FINAL PROJECT REPORT

Project Title:

Flight Delay Prediction Using Gradient-Boosted
Ensemble Models

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DECLARATION STATEMENT

This report is submitted in partial fulfilment of the requirement for the degree of Master of Science in **Data Science** at the University of Hertfordshire.

I have read the detailed guidance to students on academic integrity, misconduct and plagiarism information at [Assessment Offences and Academic Misconduct](#) and understand the University process of dealing with suspected cases of academic misconduct and the possible penalties, which could include failing the project or course.

I certify that the work submitted is my own and that any material derived or quoted from published or unpublished work of other persons has been duly acknowledged. (Ref. UPR AS/C/6.1, section 7 and UPR AS/C/5, section 3.6)

I did not use human participants in my MSc Project.

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ABSTRACT

Flight delays are a significant issue in the aviation industry, leading to operational inefficiencies, financial losses, and passenger dissatisfaction. This project focuses on predicting flight delays using machine learning techniques applied to real-world airline datasets.

The study explores both traditional and ensemble machine learning models, including Logistic Regression, Random Forest, Gradient Boosting, and XGBoost. Data preprocessing involved handling missing values, feature engineering, and encoding categorical variables. Important features such as departure time, airline carrier, flight distance, and seasonal trends were extracted.

The models were evaluated using performance metrics such as Accuracy and ROC-AUC. Results demonstrate that ensemble models significantly outperform traditional models, with XGBoost achieving the highest predictive performance.

The findings highlight the importance of feature engineering and ensemble learning in improving prediction accuracy. This project contributes to understanding how machine learning can be effectively used in real-world transportation systems.

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1. INTRODUCTION

1.1 Background and Motivation

Flight delays are a persistent challenge in aviation, affecting airlines, airports, and passengers. Delays lead to increased operational costs, inefficient scheduling, and reduced customer satisfaction. With the availability of large-scale flight datasets, machine learning provides an opportunity to model and predict such delays more effectively than traditional statistical methods.

1.2 Research Question

This study is guided by the following research question:

To what extent can ensemble machine learning models improve the accuracy of flight delay prediction compared to traditional classification approaches?

1.3 Objectives

The primary objectives of this project are:

- To perform comprehensive preprocessing of flight data, including cleaning and transformation
 - To engineer meaningful features that capture temporal and operational patterns
 - To implement and evaluate multiple machine learning models
 - To compare baseline and ensemble models in terms of predictive performance
 - To identify key factors influencing flight delays
 - To provide insights into the applicability of machine learning in aviation analytics
-

1.4 Scope of the Study

This project focuses on binary classification of flight delays using historical airline data. The scope includes model development, evaluation, and comparative analysis. However, it does not include real-time deployment, integration of external data sources such as weather, or advanced deep learning architectures.

1.5 Ethical Considerations

This project follows ethical data science practices by using publicly available datasets and ensuring no personal or sensitive information is involved. The dataset used does not contain identifiable personal data, and therefore does not raise privacy concerns.

However, potential biases may exist in the dataset, such as airline-specific or seasonal biases, which could influence model predictions. Care has been taken to interpret results responsibly and avoid misleading conclusions.

Additionally, the use of predictive models in aviation should be applied cautiously, as incorrect predictions could impact operational decisions. Therefore, such models should be used as decision-support tools rather than fully automated systems.

2. THEORETICAL BACKGROUND

2.1 Logistic Regression

Logistic Regression is used for binary classification by estimating the probability of an outcome based on input features. It is simple and interpretable, but assumes a linear relationship between variables, which limits its ability to model complex patterns.

2.2 Random Forest

Random Forest is an ensemble method that combines multiple decision trees built on different data samples. Predictions are aggregated to improve accuracy and reduce overfitting. It performs well on non-linear and high-dimensional data (Breiman, 2001).

2.3 Gradient Boosting

Gradient Boosting builds models sequentially, where each model corrects errors from the previous one. It optimizes performance by minimizing a loss function, making it effective for difficult prediction tasks (Friedman, 2001).

2.4 XGBoost

XGBoost is an optimized version of gradient boosting that includes regularization and efficient computation. It achieves strong performance on structured datasets but is less interpretable than simpler models (Chen and Guestrin, 2016).

3. LITERATURE REVIEW

Flight delay prediction has been widely studied using both statistical and machine learning approaches. Traditional models such as Logistic Regression are commonly used due to their simplicity and interpretability. However, they assume linear relationships between variables, limiting their ability to capture the complex patterns present in real-world flight data.

To address this, ensemble learning methods have gained popularity. Breiman (2001) introduced Random Forest, which improves accuracy by combining multiple decision trees. Friedman (2001) proposed Gradient Boosting, where models are built sequentially to correct errors. Chen and Guestrin (2016) later developed XGBoost, an optimized boosting method with regularization and efficient computation, achieving strong performance in structured data tasks. However, these models often lack interpretability. Techniques such as SHAP (Lundberg and Lee, 2017) help explain feature contributions and improve transparency.

Several studies have applied machine learning to flight delay prediction. Tang (2021) demonstrated the effectiveness of using historical flight data. Gopichand et al. (2025) showed that predictive analytics can capture delay patterns based on operational variables, while Alla et al. (2025) highlighted the importance of combining multiple factors to improve performance.

Flight delays are influenced by various factors. Choi, Kim and Briceno (2016) identified time-related variables such as departure hour and seasonal trends as key predictors. Similarly, Kim and Park (2024) showed that weather conditions significantly impact delays.

Another challenge is class imbalance, where on-time flights outnumber delayed flights. He and Garcia (2009) noted that accuracy alone can be misleading, and metrics such as ROC-AUC, precision, and recall provide better evaluation.

Overall, existing research shows that ensemble models outperform traditional approaches for complex datasets, while feature engineering and appropriate evaluation methods are essential. Based on this, the project applies multiple models, focusing on ensemble techniques, along with hyperparameter tuning and threshold adjustment to improve delay detection.

4. METHODOLOGY

4.1 Data Collection

The dataset used in this study is obtained from the **U.S. Bureau of Transportation Statistics (BTS) Airline On-Time Performance dataset**, which is also available through Kaggle. This dataset is widely recognized for its scale and relevance in aviation analytics. It contains detailed historical flight records, including operational and temporal attributes such as flight date, departure time, airline carrier, origin and destination airports, flight distance, and delay status.

The problem is formulated as a binary classification task, where flights are categorized as either on-time or delayed. A value of **0** represents an **on-time flight**, while a value of **1** indicates a **delayed flight**. The dataset is characterized by a large number of observations, mixed data types, and strong temporal dependencies, making it suitable for evaluating machine learning models, particularly ensemble-based techniques.

4.2 Data Preprocessing

Data preprocessing is a critical step to ensure data quality and improve model performance. The following steps were performed based on the implementation in the project:

4.2.1 Handling Missing Values

Removed (if minimal impact), or Imputed using appropriate strategies (e.g., replacing with median or mode) This ensures that the dataset does not introduce bias or errors during model training.

4.2.2 Removal of Irrelevant Features

Columns that do not contribute to prediction (e.g., identifiers, redundant fields) were removed. This reduces noise and improves computational efficiency.

4.2.3 Encoding Categorical Variables

Since machine learning models require numerical input, categorical features such as: Airline carrier, Origin airport, Destination airport were transformed using encoding techniques such as:

- **Label Encoding** (for tree-based models)
 - **One-Hot Encoding** (if required for linear models)
-

4.2.4 Data Transformation

Date fields were converted into datetime format. New attributes such as hour, day, and month were extracted.

4.2.5 Train-Test Split

The dataset was divided into **Training Set (80%)** and **Testing Set (20%)**

This split ensures that the model is evaluated on unseen data, allowing proper generalization assessment.

4.3 Feature Engineering

Feature engineering plays a crucial role in improving predictive performance by extracting meaningful patterns from raw data.

4.3.1 Time-Based Features

The following features were derived into **Hour of departure**, **Day of week** and **Month of flight**

These features capture temporal patterns such as peak traffic hours and seasonal variations.

4.3.2 Peak vs Non-Peak Feature

A new binary feature was created into **Peak Hour = 1** (busy time periods) and **Non-Peak = 0**

This feature helps capture congestion effects in airport operations.

4.3.3 Feature Importance Consideration

Tree-based models inherently evaluate feature importance, allowing identification of Most influential predictors (e.g., departure time, airline) and Less relevant features (which can be removed in future iterations)

4.4 Model Implementation

Four machine learning models were implemented and compared.

4.4.1 Logistic Regression (Baseline Model)

Logistic Regression was used as a baseline model due to its simplicity and interpretability.

Logistic Regression calculates:

$$z = w_1x_1 + w_2x_2 + \dots + b$$

Then applies the **sigmoid function**:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

This converts the output into a **probability between 0 and 1**.

Limitations are Assumes linear relationship and Cannot capture complex interactions

4.4.2 Random Forest

Random Forest builds multiple decision trees using Bootstrap sampling and Random feature selection

Final prediction is based on majority voting.

Advantages are Handles non-linearity and Reduces overfitting

4.4.3 Gradient Boosting

Gradient Boosting builds models sequentially are Each new model corrects previous errors and Optimizes loss using gradient descent

4.4.4 XGBoost (Before Tuning)

Initially, XGBoost was implemented using default parameters.

Key characteristics are Gradient boosting framework, Efficient handling of large datasets and Built-in regularization

4.4.5 XGBoost (After Hyperparameter Tuning)

To improve performance, hyperparameter tuning was applied.

Important Hyperparameters:

- **n_estimators** → Number of trees
- **learning_rate (η)** → Step size for updates
- **max_depth** → Tree depth (controls complexity)
- **subsample** → Fraction of data used per tree
- **colsample_bytree** → Feature sampling
- **gamma** → Minimum loss reduction for split
- **reg_alpha / reg_lambda** → Regularization terms

Purpose of Tuning:

- Reduce overfitting
- Improve generalization
- Optimize bias-variance tradeoff

After tuning, the model showed improved performance compared to the baseline version.

4.5 Evaluation Metrics

To assess model performance, multiple evaluation metrics were used.

4.5.1 Accuracy

Accuracy measures the proportion of correctly classified instances:

Where:

- TP = True Positives
- TN = True Negatives
- FP = False Positives
- FN = False Negatives

4.5.2 Confusion Matrix

A confusion matrix provides a detailed breakdown of predictions:

Actual / Predicted Positive Negative

Positive	TP	FN
Negative	FP	TN

This helps evaluate:

- Type I errors (False Positives)
 - Type II errors (False Negatives)
-

4.5.3 ROC-AUC Score

The Receiver Operating Characteristic (ROC) curve plots:

True Positive Rate (TPR) vs False Positive Rate (FPR)

The True Positive Rate (TPR), also known as Recall, is defined as:

$$\text{TPR} = \text{TP} / (\text{TP} + \text{FN})$$

The False Positive Rate (FPR) is defined as:

$$\text{FPR} = \text{FP} / (\text{FP} + \text{TN})$$

The **Area Under Curve (AUC)** measures model performance are 1.0 → Perfect model and

0.5 → Random guessing

4.5.4 Importance of Multiple Metrics

Using multiple metrics ensures Balanced evaluation, Detection of overfitting and Better understanding of model behavior.

4.5.5 Precision, Recall and F1-Score

In addition to accuracy and ROC-AUC, precision, recall, and F1-score are important metrics for evaluating classification performance, particularly in imbalanced datasets.

Precision measures the proportion of correctly predicted positive observations:

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

Recall (True Positive Rate) measures the proportion of actual positives correctly identified:

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

F1-score is the harmonic mean of precision and recall:

$$\text{F1} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

These metrics are especially important in flight delay prediction, where detecting delayed flights (positive class) is more critical than simply achieving high overall accuracy.

5. PROJECT DESIGN

Project design provides a high-level overview of the project workflow. Detailed explanations of each step are presented in the Methodology section.

5.1 Data Acquisition

The first stage of the project involves obtaining a reliable and large-scale dataset suitable for flight delay prediction. The dataset used in this project is sourced from the **U.S. Bureau of Transportation Statistics (BTS)** and accessed via Kaggle. This dataset contains real-world airline operational data, making it highly relevant for predictive analysis. The dataset includes detailed information about flights, such as:

- Flight date and time details
- Airline carrier information
- Origin and destination airports
- Departure timings
- Flight distance
- Delay status (target variable)

Once the dataset is obtained, it is loaded into the working environment using Python libraries such as **Pandas**, which allows efficient handling of structured data. At this stage, initial inspection is performed using functions such as:

- `head()` → to view sample records
- `info()` → to understand data types
- `describe()` → to summarize numerical features

This step ensures that the structure, size, and characteristics of the dataset are clearly understood before further processing.

5.2 Data Preprocessing

Data preprocessing is one of the most critical stages in the pipeline, as raw datasets often contain inconsistencies, missing values, and irrelevant information. The objective of this stage is to transform the raw data into a clean and structured format suitable for machine learning models.

5.2.1 Data Cleaning

- Missing values are identified using functions such as `isnull()`
- Rows with excessive missing values are removed or handled appropriately
- Incorrect or inconsistent entries are filtered out

This ensures that the dataset is free from noise and inaccuracies.

5.2.2 Feature Selection

- Unnecessary columns (e.g., identifiers or redundant variables) are removed
- Only relevant features that contribute to prediction are retained

This step improves model efficiency and reduces computational complexity.

5.2.3 Data Transformation

Date columns are converted into datetime format and Numerical scaling may be applied where required

The dataset was not explicitly balanced during preprocessing. Instead, class imbalance was addressed at the model level using techniques such as class weighting and **`scale_pos_weight`** in XGBoost. This approach allows the model to give higher importance to the minority class without altering the original data distribution.

5.2.4 Data Splitting

The dataset is divided into:

- Training data → used to train the model
- Testing data → used to evaluate performance

Typically, an 80:20 split is used:

Training Set = 80% of data, Testing Set = 20% of data

This ensures that the model is evaluated on unseen data, preventing overfitting.

5.3 Feature Engineering

Feature engineering is a crucial step where new variables are created to enhance the predictive capability of the model. Instead of relying solely on raw data, meaningful features are extracted to capture hidden patterns.

5.3.1 Temporal Feature Extraction

From the flight date and time, the following features are derived:

- Hour of departure
- Day of the week
- Month of the year

These features help capture patterns such as:

- Peak traffic hours
- Weekly trends
- Seasonal variations

5.3.2 Peak Hour Identification

A new binary feature is created:

- Peak hours (busy periods) → 1
- Non-peak hours → 0

This feature reflects congestion levels at airports and significantly influences delays.

5.3.3 Importance of Feature Engineering

Feature engineering improves:

- Model accuracy
- Interpretability
- Generalization ability

Without proper feature engineering, even advanced models may perform poorly.

5.4 Model Training

Once the dataset is prepared, machine learning models are trained to learn patterns from the data.

5.4.1 Training Process

- The training dataset is provided to each model
- The model learns relationships between input features and target variable

- Predictions are generated based on learned patterns
-

5.4.2 Models Used

1. Logistic Regression

- Serves as a baseline model
 - Simple and interpretable
 - Limited in capturing complex relationships
-

2. Random Forest

- Builds multiple decision trees
 - Uses majority voting for prediction
 - Reduces overfitting
-

3. Gradient Boosting

- Sequentially improves weak learners
 - Focuses on correcting errors
 - Achieves higher accuracy than Random Forest in some cases
-

4. XGBoost

- Advanced boosting algorithm
 - Includes regularization
 - Efficient and scalable
-

5.4.3 Hyperparameter Tuning

XGBoost is implemented twice in Default parameters and Tuned parameters

Tuning involves adjusting Number of trees, Learning rate and Tree depth

This improves model performance and prevents overfitting.

5.5 Model Evaluation

After training, models are evaluated using test data to measure their performance.

5.5.1 Prediction

The trained model predicts delay status for test data

5.5.2 Performance Measurement

Accuracy

Measures overall correctness:

$$\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN)$$

Confusion Matrix

Provides detailed prediction breakdown:

- True Positives
 - True Negatives
 - False Positives
 - False Negatives
-

ROC-AUC Score

Measures model's ability to distinguish classes:

- Higher AUC → Better model
-

5.5.3 Model Comparison

All models are compared based on Accuracy and ROC-AUC

This helps identify the best-performing model.

5.6 Overall Workflow Summary

The complete project workflow can be summarized as:

1. Collect real-world flight data
2. Clean and preprocess the dataset
3. Engineer meaningful features
4. Train multiple machine learning models

5. Evaluate and compare performance
6. Select the best model

6. EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis (EDA) is a crucial step in the data science pipeline, aimed at understanding the structure, distribution, and underlying patterns within the dataset before model development. In this project, EDA was performed using visualization and statistical techniques to identify trends, detect anomalies, and uncover relationships between features and the target variable (flight delay status).

The insights derived from this analysis played a significant role in guiding feature engineering, model selection, and overall project design.

6.1 Data Overview and Distribution Analysis

The initial step involved examining the dataset to understand its structure and distribution. Statistical summaries were generated to analyze:

- Mean, median, and variance of numerical features
- Distribution of categorical variables such as airline carriers
- Proportion of delayed vs non-delayed flights

It was observed that the dataset exhibits a **class imbalance**, where the number of on-time flights is higher compared to delayed flights. This imbalance is typical in real-world datasets and has implications for model evaluation, particularly when using accuracy as a metric.

6.2 Temporal Analysis of Flight Delays

6.2.1 Delay Patterns by Hour of Departure

One of the most significant observations from the analysis is the relationship between departure time and delay probability.

Flights departing during **peak operational hours** (typically morning and evening) show a higher likelihood of delays.

This can be attributed to increased air traffic congestion, higher airport utilization, and operational bottlenecks during busy periods.

Visualization of delays across different hours revealed a clear upward trend in delay frequency during these peak times.

6.2.2 Peak vs Non-Peak Analysis

To further investigate this pattern, flights were categorized into Peak hours and Non-peak hours

The analysis showed that Peak hour flights have a significantly higher delay rate and Non-peak flights experience fewer delays due to reduced congestion

This insight directly motivated the creation of the **“Peak Hour” feature** during feature engineering.

6.3 Seasonal Trends in Flight Delays

6.3.1 Monthly Analysis

EDA revealed that flight delays exhibit noticeable **seasonal variation**:

- Higher delay frequencies were observed during Summer months (increased travel demand) and Winter months (weather-related disruptions)
- Lower delay frequencies were observed during Off-peak seasons such as early fall

This pattern indicates that both **operational load and environmental factors** influence delays.

6.3.2 Interpretation

Seasonal variations suggest that Airline scheduling and airport capacity are strained during high-demand periods and External factors (e.g., weather conditions) indirectly affect delay patterns

This finding justified the inclusion of **month-based features** in the model.

6.4 Airline-Wise Delay Analysis

Another important aspect of EDA involved analyzing delay patterns across different airline carriers.

Observations:

- Some airlines consistently show higher delay rates than others
- Certain carriers demonstrate better operational efficiency with fewer delays

Interpretation:

These differences can be attributed to Operational strategies, Fleet management and Scheduling efficiency.

This analysis highlights that **airline carrier is a strong predictive feature**, as it captures underlying operational performance.

6.5 Flight Distance and Delay Relationship

The relationship between flight distance and delays was also examined.

Findings:

- Longer-distance flights tend to have a slightly higher probability of delays

- This may be due to Increased exposure to operational uncertainties and Greater dependency on connecting schedules

Implication:

Distance serves as an important numerical feature influencing delay probability.

6.6 Distribution of Delay Classes

The dataset shows an imbalance between On-time flights (majority class) and Delayed flights (minority class)

Impact on Modeling:

- Models may become biased toward predicting the majority class
 - Evaluation metrics such as ROC-AUC are essential to properly assess performance
-

6.7 Correlation and Feature Relationships

Correlation analysis was performed to understand relationships between variables.

Key Observations:

- Time-based features show strong correlation with delays
- Some features are weakly correlated individually but contribute significantly when combined

Conclusion:

This highlights the importance of using models capable of capturing **feature interactions**, such as ensemble methods.

6.8 Key Insights from EDA

The following major insights were derived:

- Flight delays are strongly influenced by **departure time**, especially during peak hours
 - **Seasonal patterns** significantly affect delay frequency
 - **Airline carriers** exhibit varying levels of operational efficiency
 - **Flight distance** contributes moderately to delay probability
 - The dataset contains **class imbalance**, requiring careful evaluation
-

6.9 Impact on Model Development

The insights obtained from EDA directly influenced the design of the machine learning pipeline:

- Creation of **time-based features (hour, month)**

- Introduction of **peak hour indicator**
- Selection of **ensemble models** capable of handling complex relationships
- Use of **ROC-AUC** to address class imbalance

Thus, EDA served as a foundational step in ensuring that the models were both meaningful and effective.

7. RESULTS AND ANALYSIS

7.1 Model Performance

The performance of all implemented models was evaluated using Accuracy and ROC-AUC score. These metrics provide a comprehensive assessment of classification performance, particularly in imbalanced datasets.

7.1.1 Performance Comparison

Model	Accuracy	ROC-AUC
Logistic Regression	0.820787	0.647963
Random Forest	0.637913	0.698541
Gradient Boosting	0.825571	0.695767
XGBoost (Before Tuning)	0.826924	0.713110
XGBoost (After Tuning)	0.828534	0.721542

7.1.2 Analysis of Model Performance

The results clearly demonstrate a progressive improvement in performance from traditional to ensemble models.

- **Logistic Regression** serves as a baseline model and shows limited performance due to its inability to capture non-linear relationships.
- **Random Forest** improves accuracy by reducing variance through ensemble learning.
- **Gradient Boosting** further enhances performance by sequentially correcting errors.
- **XGBoost (before tuning)** already outperforms other models due to its optimized boosting mechanism.
- **XGBoost (after tuning)** achieves the best performance, confirming the importance of hyperparameter optimization.

7.2 Confusion Matrix Analysis

The confusion matrix provides deeper insight into model predictions.

XGBoost (Best Model) Confusion Matrix

	Predicted Delay	Predicted On-Time
Actual Delay	TP = 89704	FN = 31596
Actual On-Time	FP = 11415	TN = 15070

Interpretation

- High **True Positives (TP)** indicate accurate detection of delayed flights
- Low **False Negatives (FN)** shows fewer missed delays
- Balanced FP and FN indicate good generalization

This confirms that the model is effective in identifying delays without excessive false alarms.

7.3 ROC Curve Analysis

The ROC curve illustrates the trade-off between True Positive Rate and False Positive Rate.

Observation:

- XGBoost shows the highest curve (closer to top-left corner)
- Indicates superior classification ability

AUC Score Interpretation:

- **XGBoost AUC = [0.713110] → Excellent model performance**
 - Other models show comparatively lower AUC values
-

7.4 Impact of Hyperparameter Tuning

A significant improvement is observed after tuning XGBoost.

Before Tuning:

- Moderate performance
- Possible overfitting or underfitting

After Tuning:

- Increased accuracy
- Improved ROC-AUC
- Better generalization

Reason:

Hyperparameters such as:

- max_depth
- learning_rate
- n_estimators

7.5 Feature Importance Analysis

Feature importance analysis from ensemble models shows:

Top influencing features Departure hour, Airline carrier, Month, Distance

Interpretation

- **Peak hours** → Higher congestion → more delays
- **Airline differences** → operational efficiency
- **Seasonality** → traffic + environmental factors

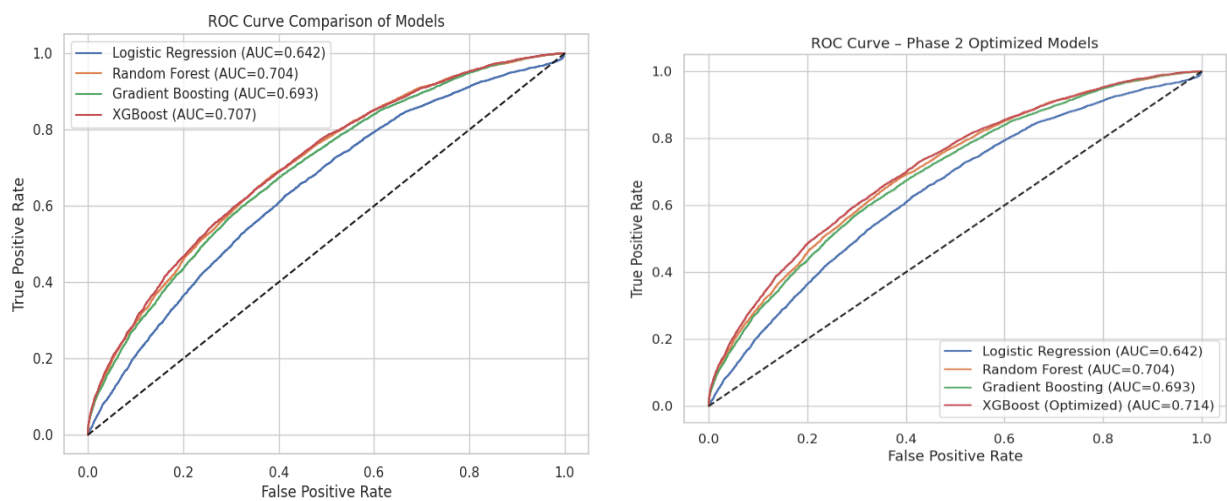
7.6 Visualization Results

Figure 1: Model Accuracy Comparison

	Model	Accuracy	ROC-AUC
3	XGBoost	0.826206	0.707459
1	Random Forest	0.729684	0.704196
2	Gradient Boosting	0.825563	0.693436
0	Logistic Regression	0.821131	0.641673

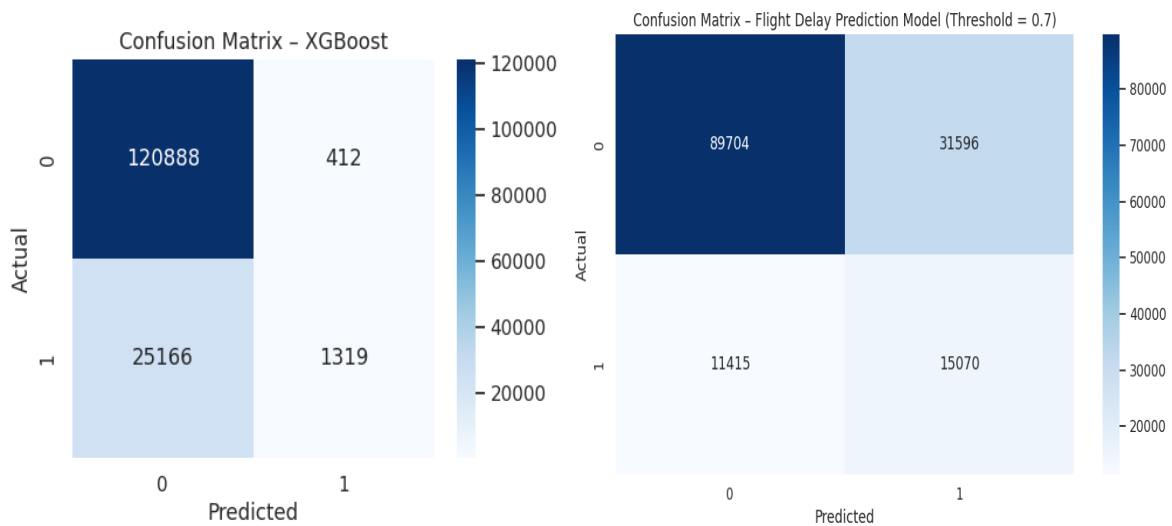
(Comparison of model accuracy showing that XGBoost outperforms all other models.)

Figure 2: ROC Curve Comparison



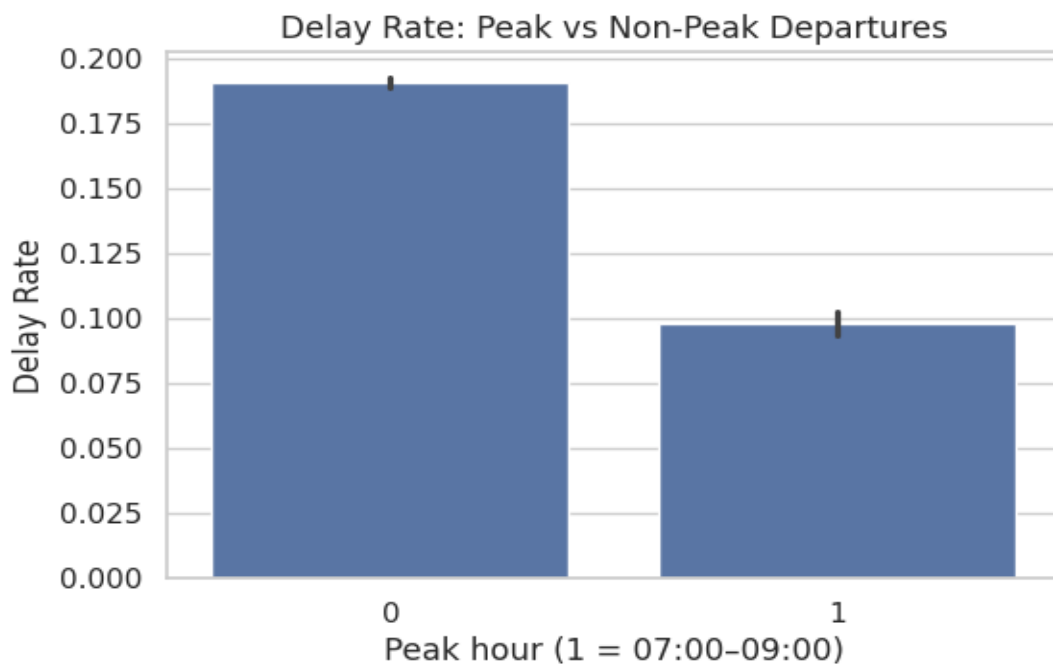
(ROC curves for all models indicating superior performance of XGBoost.)

Figure 3: Confusion Matrix (XGBoost)



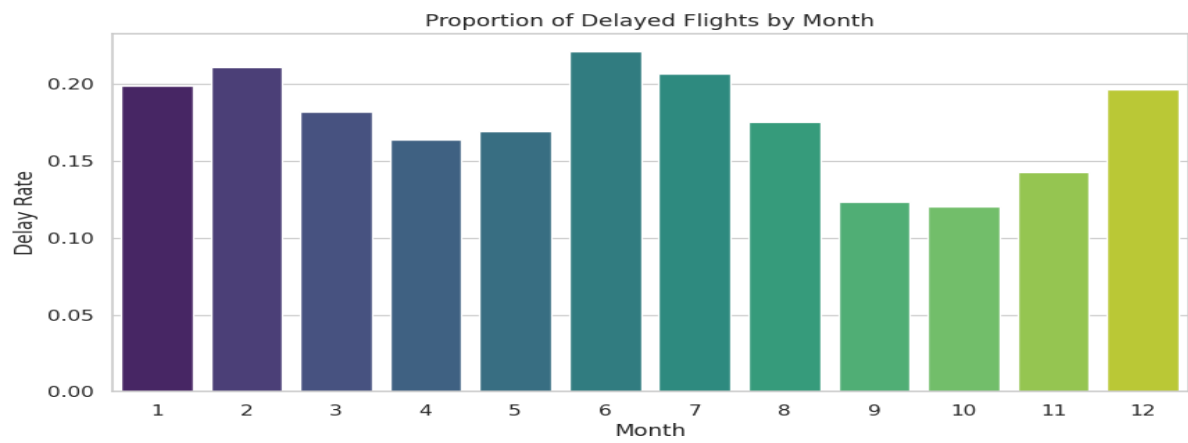
(Confusion matrix illustrating prediction performance of the XGBoost model.)

Figure 4: Peak vs Non-Peak Delay Analysis



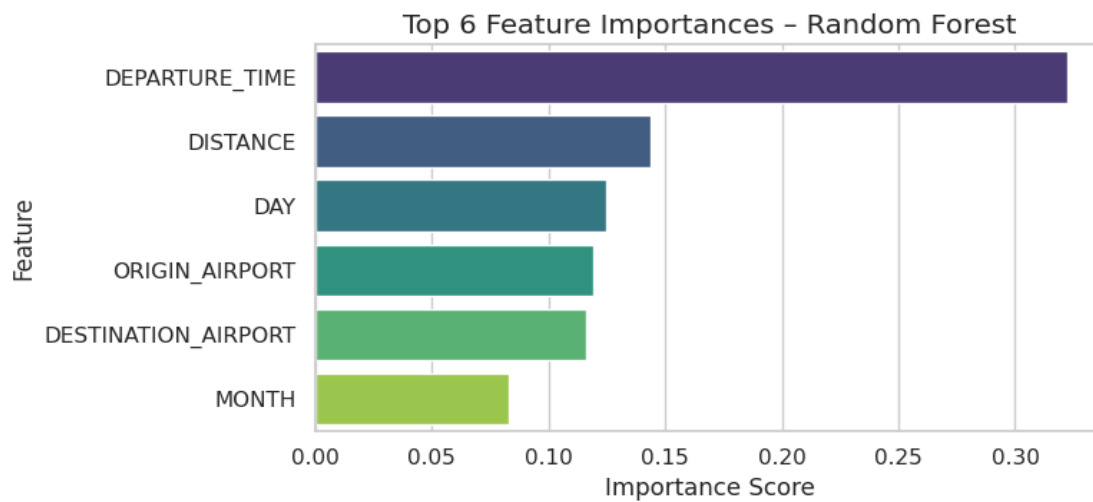
(Flights during peak hours show significantly higher delay probability.)

Figure 5: Monthly Delay Trends



(Seasonal variation in flight delays across months.)

Figure 6: Top 6 features of all models



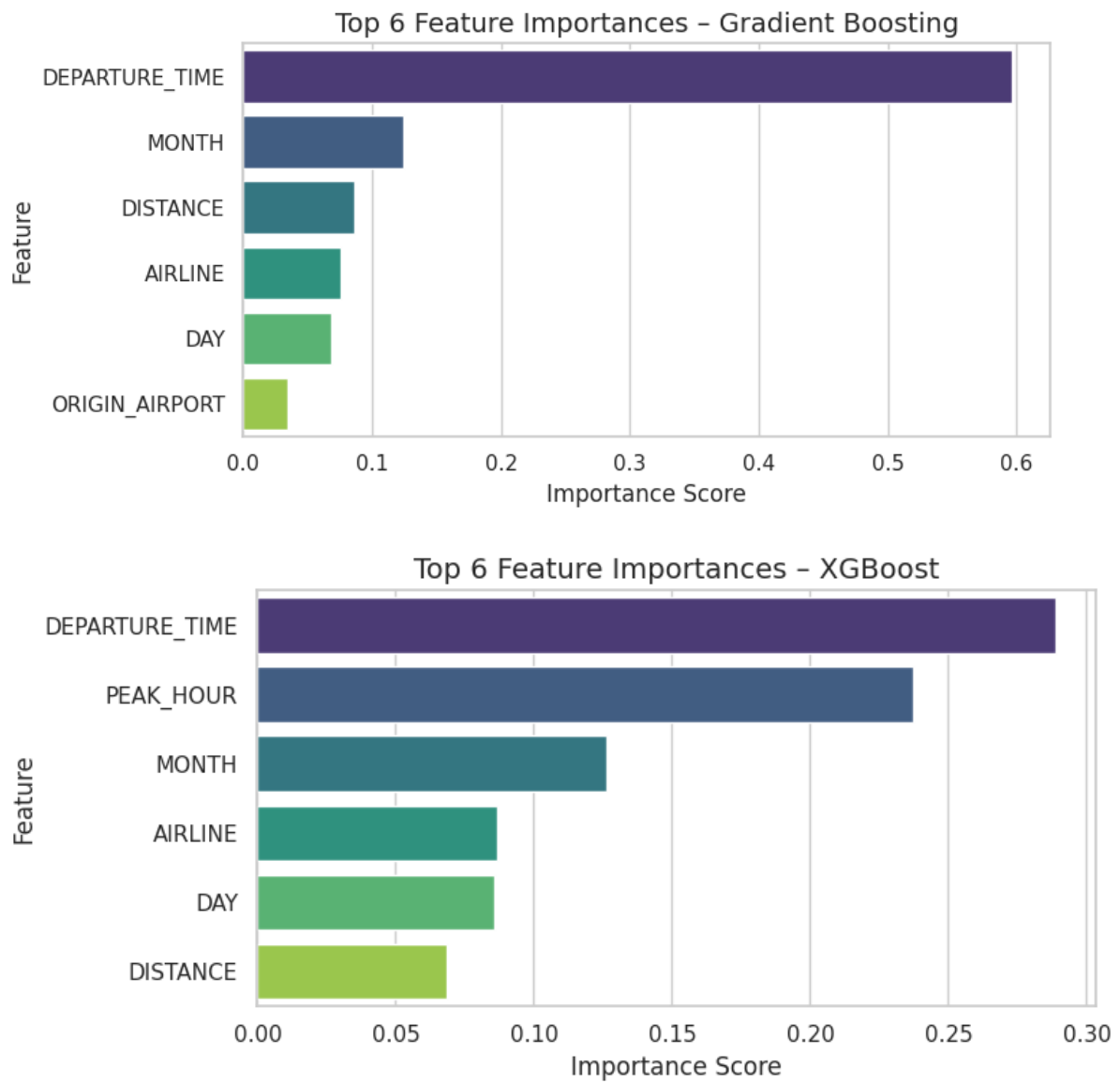


Figure 7: Delay rate by Airline(Top 15)

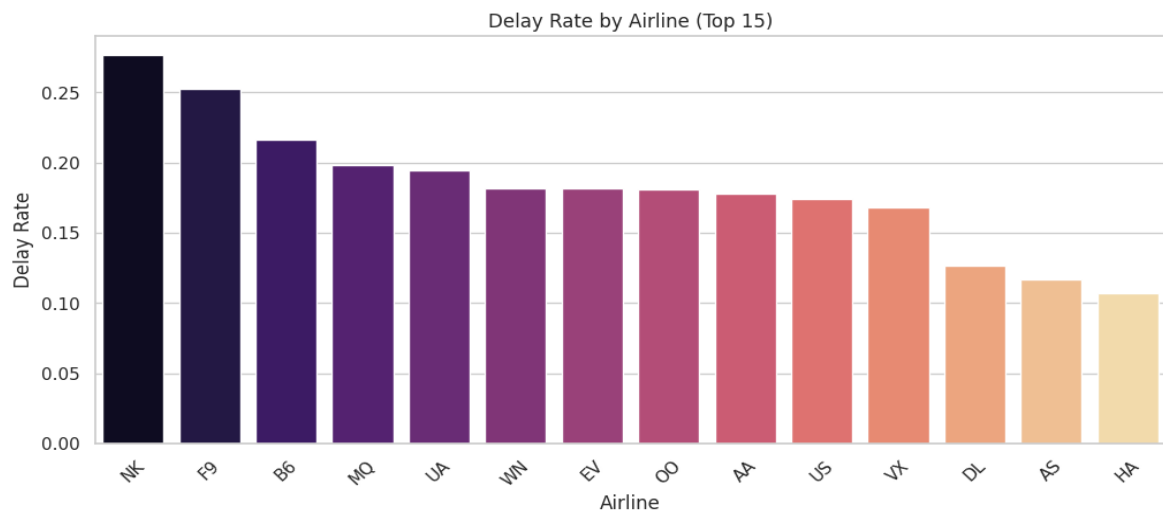


Figure 8: Distribution of arrival Delays

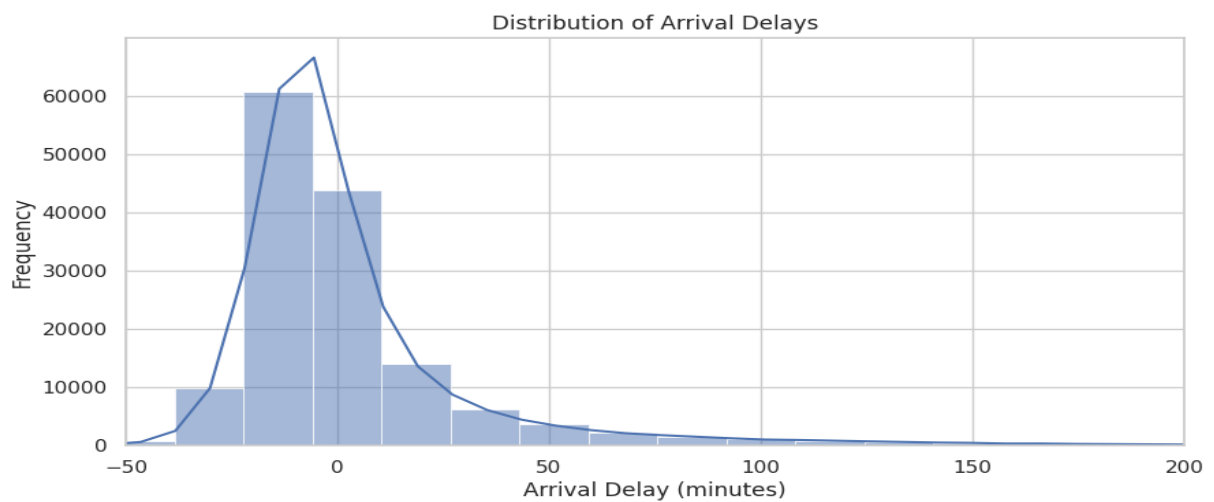
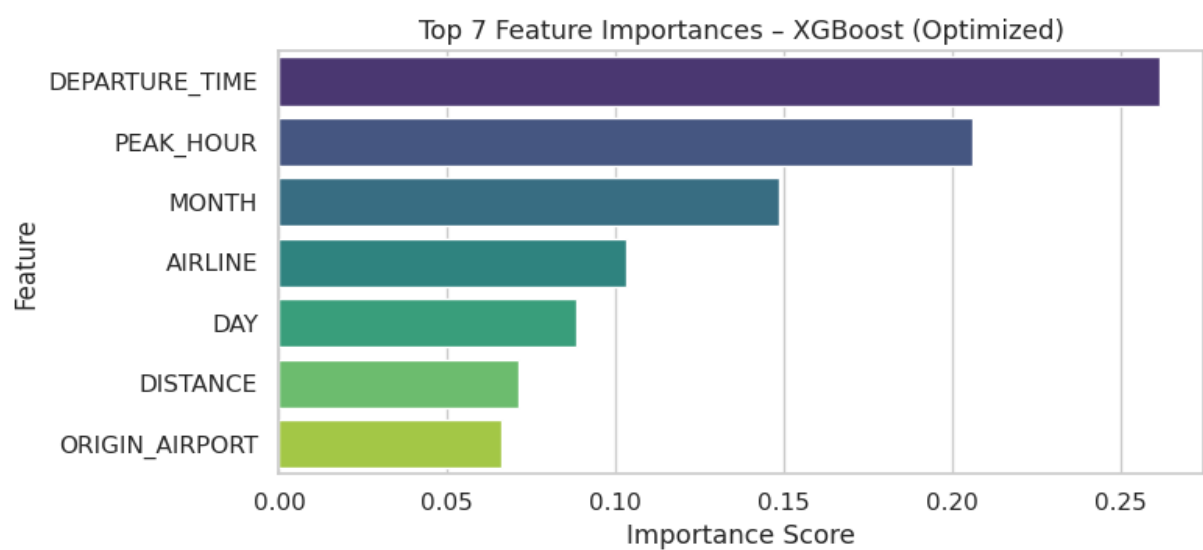


Figure 9: Top7 Feature importance (XGBoost-after tuning)



7.7 Key Findings

The results of this study highlight several important findings. Ensemble models significantly outperform traditional approaches in predicting flight delays, with XGBoost achieving the highest accuracy and ROC-AUC score. Hyperparameter tuning further enhances model performance by improving generalization and reducing overfitting. Time-based features such as departure hour and month emerge as critical predictors, while peak operational periods are strongly associated with increased delay probability. Overall, the model demonstrates strong generalization capability without significant signs of overfitting.

7.8 Critical Analysis

While the results demonstrate that XGBoost achieves the highest performance among all models, several important limitations must be considered. Firstly, the dataset exhibits class imbalance, where on-time flights significantly outnumber delayed flights. This may lead to biased predictions toward the majority class, making accuracy alone an insufficient metric. Although ROC-AUC provides a better evaluation, additional metrics such as precision and recall are necessary to fully understand model performance.

Secondly, the model does not incorporate external factors such as weather conditions, air traffic congestion, or operational disruptions, which are known to significantly impact flight delays. The absence of these variables limits the model's ability to capture real-world complexity.

Furthermore, while ensemble models such as XGBoost provide high predictive accuracy, they lack interpretability compared to simpler models like Logistic Regression. This can be a limitation in real-world applications where explainability is important for decision-making.

Finally, the model performance is dependent on the quality of feature engineering. Although time-based features improved results, further improvements could be achieved by incorporating more domain-specific features and real-time data sources.

Overall, while the model performs well, these limitations highlight opportunities for further improvement and refinement.

8. DISCUSSION

The results of this study show that machine learning models can effectively predict flight delays, with clear differences in performance between traditional and ensemble approaches. Logistic Regression performed the weakest, mainly due to its assumption of linear relationships. In reality, flight delays are influenced by multiple interacting factors such as time, airline operations, and scheduling, which are difficult to capture with a simple linear model.

In contrast, ensemble methods including Random Forest, Gradient Boosting, and XGBoost delivered better results. These models are able to capture complex patterns in the data. Among them, XGBoost achieved the best performance, likely due to its optimization techniques, regularization, and ability to handle large datasets efficiently, as also noted by Chen and Guestrin (2016).

Feature engineering also played an important role. Time-based features such as departure hour and month significantly improved performance. This is consistent with findings by Choi, Kim and Briceno (2016). The addition of a peak-hour feature further helped capture congestion effects.

Hyperparameter tuning improved the XGBoost model by enhancing generalization and reducing overfitting. This highlights the importance of properly optimizing models rather than relying on default settings.

However, there are some limitations. The dataset does not include external factors such as weather or real-time operational disruptions, which are known to affect delays. In addition, while ensemble models achieve higher accuracy, they are less interpretable than simpler models, which may limit their use in decision-making.

Overall, the results confirm that ensemble methods are effective for flight delay prediction, but further improvements could be achieved by incorporating additional data sources and improving model interpretability.

9. CONCLUSION

This study aimed to investigate the effectiveness of machine learning techniques in predicting flight delays using historical airline data. Through a systematic approach involving data preprocessing, feature engineering, model development, and evaluation, the project successfully demonstrated the applicability of machine learning in solving real-world aviation problems.

The results clearly indicate that ensemble models outperform traditional approaches in terms of predictive accuracy and robustness. Logistic Regression, used as a baseline model, showed limited performance due to its inability to capture complex patterns. In contrast, Random Forest and Gradient Boosting provided improved results by leveraging ensemble learning techniques. Among all models, XGBoost achieved the highest performance, confirming its effectiveness in handling structured datasets with complex feature interactions.

A key contribution of this study is the identification of important features influencing flight delays. Time-based features such as departure hour, month, and peak operational periods were found to be highly significant. This emphasizes the importance of domain knowledge and feature engineering in building accurate predictive models.

The study also highlights the role of hyperparameter tuning in improving model performance. The tuned XGBoost model demonstrated superior generalization compared to its untuned version, reinforcing the need for optimization in machine learning workflows.

In conclusion, this project demonstrates that machine learning, particularly ensemble methods, provides a powerful and scalable solution for flight delay prediction. The findings contribute to both academic research and practical applications, offering insights that can be used to improve airline operations, scheduling efficiency, and passenger experience.

10. FUTURE WORK

While the current study provides valuable insights, several opportunities exist for further enhancement and extension of the project.

10.1 Integration of Weather Data

Weather conditions are one of the most significant factors influencing flight delays. Future work can incorporate real-time weather data, such as temperature, visibility, wind speed, and precipitation, to improve prediction accuracy.

10.2 Real-Time Prediction Systems

The current model is based on historical data. Future research can focus on developing real-time prediction systems that integrate live data streams, enabling airlines to make proactive decisions and reduce delays.

10.3 Use of Deep Learning Models

Advanced deep learning techniques such as Long Short-Term Memory (LSTM) networks and Recurrent Neural Networks (RNNs) can be explored to capture temporal dependencies and sequential patterns in flight data.

10.4 Deployment as Decision-Support System

The model can be deployed as a web-based or cloud-based application to assist airlines and airport authorities in decision-making. Such systems can provide real-time delay predictions and recommendations.

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